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A deep learning framework for automated transaction classification and budget prediction in personal financial management

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Abstract

With the popularization of mobile payment, personal financial data has experienced explosive growth, and traditional manual recording methods can no longer meet the needs of intelligent financial management. Existing research faces challenges in processing Chinese financial texts, including insufficient semantic understanding, inadequate capture of temporal patterns, and independent processing of classification and prediction tasks. This study proposes a Deep Learning-based Financial Tracking System (DFTS) that innovatively combines the Transformer multi-head attention mechanism with Bidirectional Long Short-Term Memory (BiLSTM) network and designs a Temporal Feature Fusion Module (TFFM) to achieve collaborative optimization of transaction classification and budget prediction through a multi-task learning framework. The DFTS model employs a pre-trained Chinese BERT model to extract semantic features, introduces an amount-weighted attention mechanism to enhance perception of high-amount transactions, and captures global dependencies and multi-scale local patterns through the complementary collaboration of BiLSTM and Temporal Convolutional Network (TCN). On a self-collected Chinese dataset containing 52,847 real transaction records, the DFTS model achieves 96.3% accuracy and 95.8% F1-score in transaction classification tasks, improving by 2.2 and 2.3 percentage points compared to the strongest baseline (BERT+LSTM, achieving 94.1% accuracy and 93.5% F1-score); in budget prediction tasks, the Mean Absolute Percentage Error (MAPE) is only 6.2%, reducing by 8.6 percentage points compared to traditional ARIMA models and 3.6 percentage points compared to LSTM networks. Ablation experiments validate the effectiveness of core components including amount-weighted attention, BiLSTM, TCN, Focal Loss, and multi-task learning, while attention visualization analysis enhances model interpretability. Supplementary classification experiments on a publicly available multilingual financial transaction benchmark further confirm that the core DFTS architecture generalises effectively across language and dataset boundaries. This research provides a new technical solution for developing intelligent personal financial management systems and has significant application value in the financial technology field.



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1 Introduction

With the rapid development of mobile payment and e-commerce, personal financial data has shown explosive growth. According to statistics, more than 500 billion personal financial transaction records are generated globally every year, covering multiple life scenarios such as dining, transportation, shopping, and healthcare. How to effectively manage this massive amount of financial data and achieve automated transaction classification and accurate budget prediction has become an important research topic in the field of financial technology. Traditional personal financial management methods mainly rely on manual recording and simple rule matching [1, 2], which are not only time-consuming and labor-intensive but also difficult to cope with complex and variable consumption scenarios. In recent years, artificial intelligence technology has been increasingly widely applied in the financial field [3–5], providing new technical paths for the development of intelligent financial management systems. In particular, deep learning methods have made breakthrough progress in text classification [1, 6, 7] and time series prediction [8–10], bringing new opportunities for solving core problems in personal financial management. However, most existing research treats transaction classification and budget prediction as independent tasks, ignoring the intrinsic correlation between the two, and faces problems such as insufficient semantic understanding and inadequate capture of temporal patterns when processing Chinese financial texts.

Existing research on financial transaction classification mainly has the following shortcomings: First, traditional machine learning methods such as support vector machines and random forests [1, 2, 11], although performing reasonably well on small-scale datasets, have limited feature extraction capabilities when processing unstructured transaction description texts, with classification accuracy typically between 80 and 85%. Second, although deep learning-based methods have made some progress [6, 7, 12, 13], most adopt a single text encoder or temporal modeler, making it difficult to simultaneously capture deep semantic features of transaction descriptions and temporal dependencies of consumption behaviors. Third, research on budget prediction mainly focuses on traditional time series methods [8, 10, 14] and single recurrent neural networks [9], lacking deep mining of semantic information about user consumption habits and failing to effectively integrate multi-scale temporal features. More critically, most existing research treats classification and prediction as two independent tasks [15], ignoring the guiding role of transaction category information on budget prediction and the impact of historical consumption patterns on classification decisions. In addition, processing Chinese financial texts faces challenges such as word segmentation ambiguity, colloquial expressions, and diversity of merchant names, and the adaptability of existing methods still needs to be improved. In response to the above problems, this study proposes a Deep Learning-based Financial Tracking System (DFTS) that achieves collaborative optimization of transaction classification and budget prediction by innovatively combining the multi-head attention mechanism of Transformer with BiLSTM network and designing a temporal feature fusion module. Specifically, the main contributions of this research include:

1. A novel end-to-end multi-task learning framework DFTS is proposed, which innovatively integrates semantic encoders and temporal feature fusion modules. Through shared underlying feature representations, the transaction classification task

and budget prediction task mutually promote each other, significantly improving the overall performance of both tasks. The framework introduces an amount-weighted attention mechanism into the Transformer architecture, enabling the model to dynamically adjust the attention to different tokens according to transaction amounts, enhancing sensitivity to high-amount transactions.

2. A Temporal Feature Fusion Module (TFFM) is designed, which captures both long-term global temporal dependencies and multi-scale local consumption pattern features through the complementary collaboration of BiLSTM and Temporal Convolutional Network (TCN). The bidirectional structure of BiLSTM can model time series from both past and future directions, and the causal dilated convolution of TCN achieves efficient multi-scale feature extraction through exponentially expanded receptive fields. The fusion of the two significantly improves the accuracy of budget prediction.
3. A Chinese personal financial dataset containing 52,847 real transaction records is constructed, covering 15 main consumption categories, providing a data foundation for related research. In response to the class imbalance problem existing in the dataset (sample size difference reaches 49 times), Focal Loss is introduced to dynamically adjust sample weights, effectively improving the model's classification performance on long-tail categories.
4. The superiority of the DFTS model is comprehensively verified through extensive comparative experiments, ablation experiments, hyperparameter sensitivity analysis, and attention visualization. To further assess architectural robustness, supplementary classification experiments are conducted on a publicly available multilingual financial transaction benchmark [3]. These experiments confirm that the shared semantic-temporal backbone (Transformer encoder, BiLSTM, and TCN) transfers competitively to English-language data; it should be noted that the DFTS-specific contributions (amount-weighted attention, multi-task learning, and budget prediction) are necessarily disabled in this setting due to the absence of amount and temporal supervision signals in the benchmark, and therefore this evaluation speaks to backbone generalizability rather than to DFTS as a whole. The dataset and implementation code are available upon reasonable request to the corresponding author.

The organizational structure of this paper is as follows: Section 2 systematically reviews related research on financial transaction classification, budget prediction technology, and deep learning applications in the financial field; Section 3 details the architecture design of the DFTS model, including embedding layer, semantic encoder, temporal feature fusion module, dual output heads, and loss function and training strategy; Section 4 presents experimental results including experimental setup, dataset description, baseline method comparison, ablation experiments, hyperparameter sensitivity analysis, and supplementary backbone generalization evaluation on a public English-language benchmark (Sect. 4.9), which assesses the transferability of the shared Transformer–BiLSTM–TCN components independently of the DFTS-specific contributions; Section 5 discusses the advantages, limitations, and future research directions of the DFTS model; Section 6 summarizes the full text and looks forward to application prospects.

2 Related work

2.1 Financial transaction classification methods

Automatic financial transaction classification is one of the core tasks in personal financial management and enterprise financial analysis, aiming to automatically map transaction description texts to predefined category spaces. Early research mainly adopted rule-based methods and traditional machine learning algorithms. García-Méndez et al. [1] proposed a short text classification method based on Support Vector Machine (SVM), using TF-IDF feature extraction technology to construct a specialized labeled corpus, achieving good classification results on banking transaction data; this dataset is also adopted as a cross-dataset generalization reference in Sect. 4.9 of the present study. However, this method relies on manual feature engineering and is difficult to capture deep semantic information in texts. Patil and Dharwadkar [2] used ensemble learning methods such as random forests to analyze banking data, which although performing well when processing structured features, have limited modeling capabilities for unstructured texts. To address the class imbalance problem, Tanha et al. [16] systematically evaluated the performance of various Boosting methods in multi-class imbalanced data classification, providing important references for subsequent research.

With the development of deep learning technology, neural network-based methods have gradually become mainstream. Jørgensen and Igel [6] adopted character-level word embedding technology for cross-company financial transaction classification, effectively handling the diversity and spelling variations of merchant names by capturing character-level language patterns. Sukumaran [7] proposed an improved customer transaction classification method based on semi-supervised knowledge distillation, utilizing a large amount of unlabeled data to improve model generalization ability. Dayioglugil and Akgul [12] designed continuous embedding spaces for bank transaction data, capturing semantic associations between transactions by learning low-dimensional dense representations. Khazane et al. [13] proposed the DeepTrax model, using graph embedding technology to represent financial transactions as graph structures, capturing complex relationship networks between transactions. These deep learning-based methods significantly improve classification performance, but most focus on single text encoding or graph structure modeling, failing to fully integrate multi-modal information and temporal dependencies. In addition, existing research mainly targets English transaction data, with relatively little research on processing Chinese financial texts, while Chinese texts have unique challenges in word segmentation, semantic understanding, and colloquial expressions.

2.2 Budget prediction and time series forecasting technology

Budget prediction aims to predict future expenditure trends based on historical consumption data and is an important part of personal financial planning and enterprise capital management. Traditional time series forecasting methods such as ARIMA can model linear dependencies under the assumption of data stationarity but are difficult to capture nonlinear and variable consumption patterns. Parmezan et al. [14] systematically evaluated the performance of statistical models and machine learning models in time series prediction, pointing out the applicable conditions and limitations of different models. Schreiber and Schmitz [17] proposed surrogate time series methods that

provide a new perspective for time series data analysis but require substantial computational resources in practical applications.

Deep learning methods have made significant progress in the field of time series prediction. Salinas et al. [8] proposed the DeepAR model, which uses autoregressive recurrent networks to achieve probabilistic forecasting, capable of simultaneously predicting multiple related time series and quantifying prediction uncertainty. Sherstinsky [9] systematically expounded the basic principles of Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks, providing a theoretical foundation for temporal modeling. Alexandrov et al. [10] developed the GluonTS framework that integrates various probabilistic time series models, providing researchers with a unified experimental platform. Lai et al. [18] proposed the LSTNet model that simultaneously captures long-term and short-term temporal patterns through a combination of convolutional and recurrent layers, performing excellently on multiple time series prediction tasks. However, these methods mainly focus on numerical time series and fail to effectively utilize semantic information to assist prediction for financial transaction data containing rich text descriptions. In addition, existing temporal models mostly adopt a single modeling strategy, making it difficult to simultaneously capture global dependencies and multi-scale local patterns.

Recent research has begun to focus on time series data augmentation techniques to improve model performance. Wen et al. [19] comprehensively reviewed time series data augmentation methods for deep learning, summarizing various data augmentation strategies and their applicable scenarios. These techniques provide new ideas for solving time series data sparsity and improving model generalization ability. In terms of multi-scale temporal feature learning, although some research has explored temporal patterns at different granularities [20], how to effectively integrate multi-scale features remains an open problem, especially in the application scenario of personal financial prediction, which needs to simultaneously consider consumption patterns at different time scales such as daily, weekly, and monthly.

2.3 AI applications in finance and multi-task learning

Artificial intelligence technology is increasingly widely applied in the financial field, from customer behavior analysis to risk prediction, from robo-advisors to anti-fraud detection, and deep learning methods are profoundly changing the form of financial services. Soni et al. [3] reviewed the entire process of artificial intelligence in the business field from research innovation to market deployment, pointing out the enormous potential of AI technology in the financial technology field. Tiwari et al. [4] specifically investigated the application of artificial intelligence technology in finance and marketing, emphasizing the value of machine learning algorithms in scenarios such as customer segmentation, credit scoring, and personalized recommendation. Kaur et al. [5] discussed the impact of artificial intelligence on the banking industry in the Banking 4.0 era, analyzing how AI reshapes the business models and service processes of modern banks. Dawood et al. [11] improved bank customer behavior profiling using machine learning methods, identifying different customer groups and predicting their consumption preferences through clustering and classification algorithms. These studies indicate that AI technology has broad application prospects in the financial field, but in the subdivided field of

personal financial management, there is still a lack of end-to-end intelligent systems for Chinese users.

Multi-task learning, as an effective machine learning paradigm, can improve model generalization ability and reduce overfitting risk by sharing knowledge among multiple related tasks. Kotios et al. [15] proposed a personalized financial management system for small and medium-sized enterprises, adopting a multi-task learning framework to jointly optimize multiple financial analysis tasks, validating the effectiveness of inter-task knowledge sharing. In the medical and health field, multi-task learning has also achieved significant results. Yuan et al. [21] proposed a dynamic weighted ensemble learning framework that achieves cardiovascular risk prediction through SHAP interpretability analysis, Zhang et al. [22] designed a structure-preserving spectral graph neural network that performs excellently in gastric cancer drug synergy scoring, and Liu et al. [20] proposed a multi-scale temporal deep learning framework MSTDP that adopts a cross-attention fusion mechanism to achieve software defect prediction. These studies provide important inspiration for the multi-task learning framework design in this paper. In addition, the decentralized neural network framework SwarmSense-DNN proposed by Yang et al. [23] and the perception uncertainty quantification method proposed by Tian et al. [24] demonstrate the potential of deep learning in handling complex real-time systems, providing references for the robustness design of the DFTS model.

In terms of pre-trained language models, BERT and its variants have made breakthrough progress in Chinese natural language processing tasks. Cui et al. [25] proposed a whole word masking Chinese BERT model that significantly improves Chinese text understanding ability through improved pre-training strategies, laying the foundation for the semantic encoder design in this research. In terms of loss function design, Lin et al. [26] proposed Focal Loss that effectively solves the class imbalance problem in object detection by re-weighting difficult samples, and this method has been widely applied to various classification tasks. Kim [27] proposed a Convolutional Neural Network (CNN) text classification method, which although performing excellently in sentence-level classification, has limitations in capturing long-distance dependencies. These foundational studies provide solid theoretical support for the technical selection and innovative design of the DFTS model.

In summary, existing research has made important progress in financial transaction classification, time series prediction, and multi-task learning, but still has the following shortcomings: (1) Most existing classification methods target single-language (mainly English) scenarios and lack deep optimization for Chinese financial texts, and cross-dataset generalizability across language boundaries is rarely validated; (2) Budget prediction research mainly relies on pure numerical temporal models and fails to fully mine semantic information in transaction descriptions; (3) Classification and prediction tasks are usually processed independently, ignoring the intrinsic correlation and complementarity between them; (4) Temporal feature extraction methods are single, making it difficult to simultaneously capture global dependencies and multi-scale local patterns. The DFTS model proposed in this study effectively solves the above problems by innovatively integrating Transformer multi-head attention mechanism, BiLSTM temporal modeling, and TCN multi-scale feature extraction, and adopting a multi-task learning framework to collaboratively optimize classification and prediction tasks, providing a new technical solution for the development of intelligent personal financial management systems.

3 Methods

3.1 Problem definition

Given a dataset $D = \{(x_i, y_i, t_i, a_i)\}_{i=1}^N$ containing N transaction records, where x_i represents the text description of the i -th transaction, $y_i \in \{1, 2, \dots, C\}$ is the corresponding category label (C is the total number of categories), t_i is the timestamp, and a_i is the transaction amount. This study needs to solve two core tasks: (1) Transaction classification task, learning a mapping function $f: x \rightarrow y$ to automatically classify transaction description x into the correct category y ; (2) Budget prediction task, predicting the total expenditure within a future time window W based on historical transaction sequences.

3.2 DFTS model architecture

The DFTS model adopts an end-to-end multi-task learning architecture, achieving collaborative optimization of transaction classification and budget prediction through deep fusion of semantic features and temporal features. As shown in Fig. 1, the entire system contains five core modules: input data processing, embedding layer, semantic encoder, Temporal Feature Fusion Module (TFFM), and dual output heads. The input data includes three dimensions of information: Chinese transaction description text, transaction amount, and timestamp. The embedding layer is responsible for converting multi-modal inputs into unified vector representations, where text is extracted by a pre-trained BERT encoder to extract semantic features, Positional Encoding (PE) preserves sequence order information, and Amount Encoding (AE) maps numerical amounts to the embedding space. The semantic encoder is based on the Transformer architecture, adopting a 6-layer stacked multi-head self-attention mechanism, with each layer containing 8 attention heads, capable of capturing semantic associations at different granularities in transaction descriptions in parallel.

The Temporal Feature Fusion Module is the innovative core of DFTS, achieving comprehensive modeling of consumption patterns through the complementary collaboration of BiLSTM network and Temporal Convolutional Network (TCN). The BiLSTM network adopts a bidirectional structure with a hidden layer dimension of 256, where the forward layer captures temporal dependencies from history to present, and the backward layer extracts contextual information from future to present, and the two are concatenated to

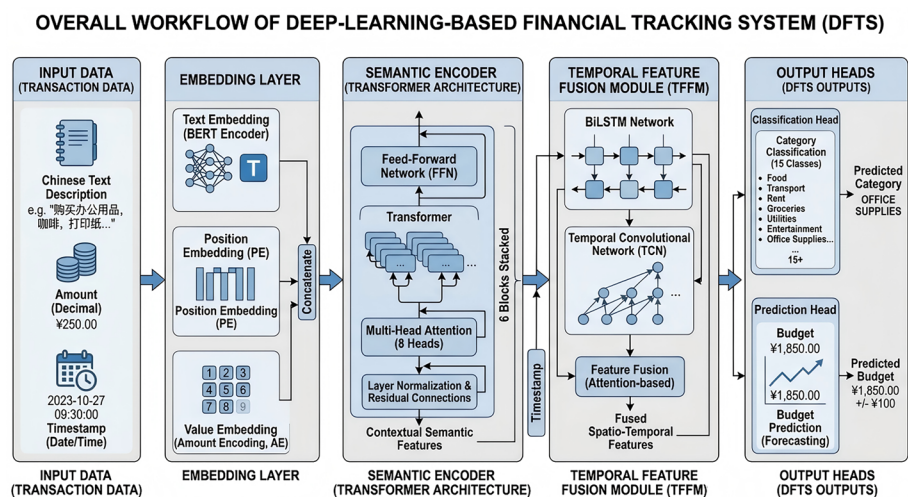


Fig. 1 Overall workflow of the DFTS model

form a complete temporal representation. TCN adopts causal dilated convolution with dilation factors set to [1, 2, 4, 8] and kernel size of 3, achieving multi-scale temporal feature extraction through exponentially expanded receptive fields, which not only ensures causality of prediction tasks but also greatly reduces computational complexity. Finally, through an attention-based feature fusion mechanism, the global dependencies of BiLSTM and local patterns of TCN are adaptively integrated to output fused spatio-temporal feature representations. The output layer contains two branches: classification head and prediction head. The classification head outputs probability distributions of 15 categories through fully connected layers and Softmax activation function; the prediction head aggregates sequence features through attention pooling mechanism and obtains budget prediction values through regression of two fully connected networks. Figure 2 shows the detailed network structure of the model, including dimension annotations for each layer, data flow directions, and residual connection designs.

3.3 Embedding layer

The embedding layer is responsible for converting heterogeneous multi-modal inputs into unified dense vector representations. For an input transaction sequence $x = [w_1, w_2, \dots, w_L]$ of length L , we adopt a pre-trained Chinese BERT model as the basis of the text encoder and integrate positional information and amount information. The output of the embedding layer is calculated as shown in Eq. (1):

$$E = \text{BERT}(x) + \text{PE} + \text{AE} \quad (1)$$

where $\text{BERT}(x) \in \mathbb{R}^{L \times d}$ is the word vector matrix encoded by BERT, and d is the embedding dimension (in this study $d = 768$). The positional encoding PE is generated using sine-cosine functions, and its calculation formula is:

$$\begin{aligned} \text{PE}_{\text{pos}, 2i} &= \sin(\text{pos}/10000^{2i/d}) \\ \text{PE}_{\text{pos}, 2i+1} &= \cos(\text{pos}/10000^{2i/d}) \end{aligned} \quad (2)$$

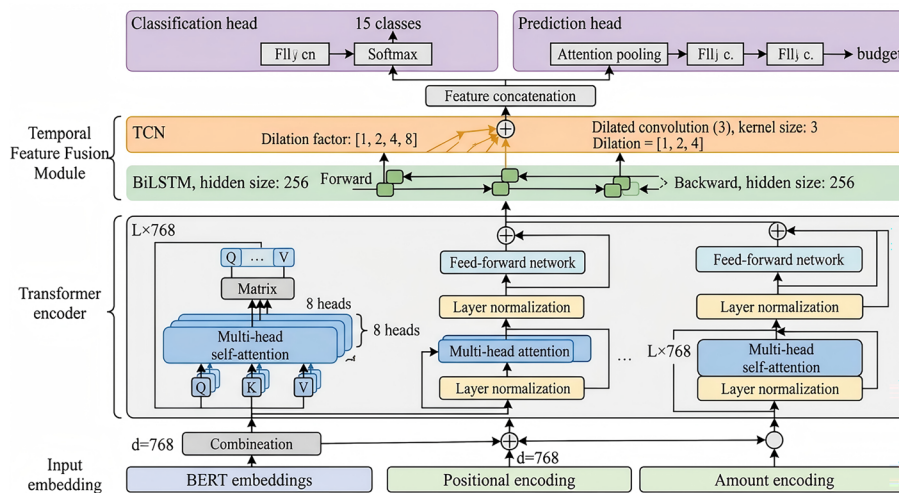


Fig. 2 Detailed network architecture of the DFTS model

The amount encoding AE maps the transaction amount a to the embedding space through logarithmic transformation and fully connected layers to capture the scale characteristics of amounts. The calculation formula for amount encoding is:

$$AE = W_a \cdot \text{ReLU}(\log(1 + a)) + b_a \quad (3)$$

where $W_a \in \mathbb{R}^{d \times 1}$ and $b_a \in \mathbb{R}^d$ are learnable weight parameters and bias terms. The logarithmic transformation can effectively compress the numerical range of amounts, enabling the model to maintain stable learning ability for transaction amounts of different magnitudes.

3.4 Semantic encoder

The semantic encoder is based on the Transformer architecture, adopting a 6-layer stacked multi-head self-attention module, with each layer containing 8 attention heads. The multi-head attention mechanism can learn dependencies of input sequences in different representation subspaces in parallel, and its calculation process is shown in Eq. (4):

$$\begin{aligned} \text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \\ \text{head}_i &= \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \end{aligned} \quad (4)$$

where h is the number of attention heads ($h = 8$), Q, K, V are the query, key, and value matrices respectively, and W_i^Q, W_i^K, W_i^V, W^O are learnable projection matrices. The calculation of single-head attention mechanism adopts scaled dot-product attention, and its formula is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (5)$$

where $d_k = d/h$ is the dimension of each attention head, and the scaling factor $\sqrt{d_k}$ is used to prevent gradient vanishing caused by excessively large dot product values. To enhance the model's perception ability of transaction amounts, we introduce an amount weighting term α in the attention mechanism, and the modified attention score calculation is shown in Eq. (6):

$$\text{Attention}'(Q, K, V) = \text{softmax}\left(\frac{QK^T + \alpha \odot M}{\sqrt{d_k}}\right) V \quad (6)$$

Here $\alpha \in \mathbb{R}$ is a globally shared learnable scalar that scales the magnitude of the amount bias, and $M \in \mathbb{R}^{L \times L}$ is the amount mask matrix constructed by broadcasting the log-transformed transaction amount across all attention positions:

$$M_{ij} = \log(1 + a), \quad \forall i, j \in \{1, \dots, L\}, \quad (7)$$

where a is the scalar transaction amount associated with the current input. The symbol $\odot$ denotes element-wise multiplication. An additive formulation is adopted rather than a multiplicative gating scheme because it preserves unobstructed gradient flow through the original QK^T term and avoids the numerical instability that multiplicative interactions may introduce in the early stages of training when α is randomly initialised. Intuitively, for a high-amount transaction, the positive bias $\alpha \odot M$ uniformly raises all

pairwise attention logits before the softmax, causing the model to attend more broadly to contextual tokens and thereby extract richer semantic representations for transactions whose amounts signal financial significance. This mechanism enables the model to dynamically adjust the attention distribution according to the transaction amount size, assigning higher attention to high-amount transactions. Each Transformer layer also contains Feed-Forward Neural Networks (FFN) and residual connections, and the output is standardized through Layer Normalization to ensure training stability.

3.5 Temporal feature fusion module

The Temporal Feature Fusion Module (TFFM) achieves comprehensive capture of consumption temporal patterns through the synergistic effect of BiLSTM and TCN. The BiLSTM network adopts a bidirectional structure, where the forward LSTM models temporal dependencies from past to present, and the backward LSTM extracts contextual information from future to past. At time t , the hidden state calculation of BiLSTM is shown in Eq. (8):

$$\begin{aligned}\vec{h}_t &= \text{LSTM}_f(E_t, \vec{h}_{t-1}) \\ \overleftarrow{h}_t &= \text{LSTM}_b(E_t, \overleftarrow{h}_{t+1}) \\ h_{\text{BiLSTM}} &= [\vec{h}_t; \overleftarrow{h}_t]\end{aligned}\quad (8)$$

where $[\cdot]$ denotes the vector concatenation operation. TCN adopts a causal dilated convolution structure, achieving efficient multi-scale feature extraction through exponentially expanded receptive fields. For the input sequence $H = \{h_{\text{BiLSTM}}\}_{t=1}^T$, the output of TCN at time t is shown in Eq. (9):

$$F_t = \sum_{i=0}^{k-1} W_i \cdot H_{t-d \cdot i} \quad (9)$$

where k is the kernel size ($k = 3$), d is the dilation factor, and W_i is the kernel weight. This study adopts a multi-layer stacked TCN structure with dilation factors set to $[1, 2, 4, 8]$. For a stack of L dilated-convolution layers with kernel size k and dilation factors $d_1, d_2, \dots, d_L$, the total receptive field is given by:

$$\text{RF} = 1 + (k - 1) \sum_{l=1}^L d_l. \quad (10)$$

Applying this formula layer by layer with $k = 3$ and $[d_1, d_2, d_3, d_4] = [1, 2, 4, 8]$ yields:

$$\begin{aligned}\text{RF}_1 &= 1 + (3 - 1) \times 1 = 3, \\ \text{RF}_2 &= \text{RF}_1 + (3 - 1) \times 2 = 7, \\ \text{RF}_3 &= \text{RF}_2 + (3 - 1) \times 4 = 15, \\ \text{RF}_4 &= \text{RF}_3 + (3 - 1) \times 8 = 31.\end{aligned}$$

The four-layer stack therefore achieves a total receptive field of

$$\text{RF} = 1 + (3 - 1) \times (1 + 2 + 4 + 8) = 31 \text{ time steps}, \quad (11)$$

effectively capturing monthly-level consumption cycles. Finally, through residual connections and layer normalization, the global features of BiLSTM and local

features of TCN are fused to output a unified spatio-temporal feature representation $Z = \text{LayerNorm}(h_{\text{BiLSTM}} + F)$.

3.6 Classification head and prediction head

The DFTS model adopts a dual-branch output structure to simultaneously complete transaction classification and budget prediction tasks. The classification head maps the spatio-temporal features Z to a C -dimensional category space through fully connected layers and uses the Softmax activation function to output probability distributions, as shown in Eq. (12):

$$p(y|x) = \text{softmax}(W_{\text{cls}}Z + b_{\text{cls}}) \quad (12)$$

where $W_{\text{cls}} \in \mathbb{R}^{C \times d}$ and $b_{\text{cls}} \in \mathbb{R}^C$ are the weights and biases of the classification layer. The prediction head first aggregates sequence features through an attention pooling mechanism, calculates the importance weights of features at each time step, and then obtains a global representation through weighted summation. After attention pooling, budget prediction values are obtained through regression of two fully connected networks (FC1 and FC2), and the calculation process is $\hat{B} = \text{FC}_2(\text{ReLU}(\text{FC}_1(\text{AttentionPooling}(Z))))$. The dual-branch structure enables the classification task and prediction task to share underlying semantic and temporal feature representations, improving model generalization ability through multi-task learning.

3.7 Loss function and training strategy

The DFTS model adopts a multi-task learning framework, achieving collaborative improvement of two tasks through joint optimization of classification loss and prediction loss. The total loss function is defined as the weighted sum of classification loss and prediction loss, as shown in Eq. (13):

$$L_{\text{total}} = \lambda_1 L_{\text{cls}} + \lambda_2 L_{\text{pred}} \quad (13)$$

where λ_1 and λ_2 are task weight hyperparameters, and this study determines the optimal values as $\lambda_1 = 0.6$ and $\lambda_2 = 0.4$ through grid search; specifically, λ_1 was swept over $\{0.1, 0.2, \dots, 0.9\}$ with $\lambda_2 = 1 - \lambda_1$, and the pair that minimised the combined validation loss was selected. The resulting $\lambda_1 > \lambda_2$ reflects the primary emphasis placed on classification accuracy while still allowing the prediction objective to regularise the shared feature representations and prevent task-specific overfitting. The prediction loss L_{pred} uses Mean Squared Error (MSE) to calculate the difference between predicted and true values, i.e., $L_{\text{pred}} = (\hat{B} - B)^2$. To effectively address the class imbalance problem existing in the dataset (as shown in Table 1, the category sample size difference reaches 50 times), the classification loss L_{cls} adopts Focal Loss, which makes the model pay more attention to difficult-to-classify samples by dynamically adjusting the weights of different samples. The calculation formula of Focal Loss is $L_{\text{cls}} = -\sum_{c=1}^C \alpha_c (1 - p_c)^\gamma \log(p_c)$, where α_c is the balance weight of category c , and γ is the focusing parameter (set to $\gamma = 2.0$ in experiments). The model training uses the AdamW optimizer with an initial learning rate of 2×10^{-5} , a batch size of 32, and training for 50 epochs. A Cosine Annealing learning rate scheduling strategy is adopted, causing the learning rate to decay periodically during the training process, which helps the model escape local optima. At the same time, an Early Stopping mechanism is introduced, which terminates training

Table 1 Distribution statistics of dataset categories

Category	Number of transactions	Percentage (%)	Average amount (¥)
Food & dining	12,456	23.6	48.30
Transportation	8923	16.9	32.50
Shopping	7845	14.8	215.80
Entertainment	5234	9.9	127.40
Utilities	3678	7.0	285.60
Healthcare	2912	5.5	368.90
Education	2456	4.6	542.30
Travel	2134	4.0	1235.70
Housing	1890	3.6	2450.00
Communication	1567	3.0	128.50
Personal care	1234	2.3	186.20
Insurance	987	1.9	1850.00
Gifts & donations	823	1.6	328.40
Investment	456	0.9	5680.00
Others	252	0.5	142.70
Total	52,847	100.0 ^a	362.00 ^b

^aPercentages are independently rounded to one decimal place and may not sum to exactly 100.0. ^bWeighted average computed as $\sum_i (N_i \times \bar{a}_i) / 52,847 = 19,128,825 / 52,847 \approx 362.00$ ¥

when the validation set performance does not improve for 5 consecutive epochs, effectively preventing overfitting.

4 Experimental results

4.1 Dataset

The dataset used in this study is derived from the author's self-collected personal financial transaction records from January 2021 to December 2024, covering multiple payment channels such as Alipay, WeChat Pay, and bank cards. The dataset contains 52,847 transaction records involving 15 main consumption categories. Each record includes information such as transaction date, transaction description, amount, and category. To prevent data leakage inherent in temporal data, the dataset is partitioned *chronologically* rather than randomly: transactions from January 2021 to June 2023 (approximately 70% of all records) constitute the training set (37,000 records), transactions from July 2023 to March 2024 form the validation set (7924 records), and transactions from April 2024 to December 2024 serve as the test set (7923 records). This strict temporal ordering ensures that the model is always evaluated on future transactions it has never observed during training, faithfully reflecting realistic deployment conditions and avoiding the inflated accuracy that random splitting would produce for a single-user dataset with recurring merchants. To comprehensively demonstrate the distribution characteristics of the dataset, Table 1 statistics the sample size, proportion, and average transaction amount of each category. It can be observed that the dataset exhibits obvious class imbalance phenomenon. The category with the most samples, "Food & Dining" (12,456 records), is 49 times the category with the least samples, "Others" (252 records), which poses a challenge to the classification performance of the model.

4.2 Experimental setup

All experiments were conducted on a server equipped with NVIDIA RTX 4090 GPU (24GB memory) using the PyTorch 2.0 deep learning framework. The implementation code is made available upon reasonable request to the corresponding author. To ensure

the reproducibility of experimental results, Table 2 details all hyperparameter settings of the DFTS model, including network structure parameters, training parameters, and loss function parameters. Among them, the embedding dimension is set to 768 to match the output dimension of the pre-trained BERT model, the number of Transformer layers and attention heads are set to 6 and 8 respectively, the BiLSTM hidden layer size is 256, and TCN adopts a multi-scale convolution structure with dilation factors [1, 2, 4, 8]. The optimizer uses AdamW, the learning rate is 2×10^{-5} , the batch size is 32, and training for 50 epochs. The loss weights λ_1 and λ_2 for multi-task learning are set to 0.6 and 0.4 respectively, and the focusing parameter γ of Focal Loss is set to 2.0.

4.3 Baseline methods

To comprehensively evaluate the performance of the DFTS model, we selected 7 representative baseline methods for comparative experiments, including traditional machine learning methods and deep learning methods. Traditional methods include: (1) SVM+TF-IDF, using term frequency-inverse document frequency features and support vector machine classifier; (2) Random Forest, a random forest ensemble learning method based on bag-of-words model. Deep learning methods include: (3) TextCNN, the text convolutional neural network proposed by Kim [27]; (4) BiLSTM, the standard bidirectional long short-term memory network; (5) BERT-base, a classifier fine-tuned based on the pre-trained BERT model; (6) BERT+LSTM, a hybrid model combining BERT encoder and LSTM temporal modeling; (7) Transformer-base, a baseline model adopting standard Transformer architecture. All baseline methods were trained and evaluated on the same chronologically partitioned training, validation, and test sets, with hyperparameters optimized on the validation set through grid search.

To ensure a fair and unconfounded comparison, the following implementation conventions are applied uniformly across all methods. (i) *Pre-trained encoder*. All BERT-based methods (BERT-base, BERT+LSTM, and Transformer-base) use the same Chinese whole-word-masking BERT checkpoint (hfl/chinese-bert-wwm-ext; Cui et al. [25]) as DFTS, so that any performance differences reflect architectural choices rather than the encoder. (ii) *Loss function*. All deep learning baselines are trained with Focal Loss ($\gamma = 2.0$, same class-balance weights α_c as DFTS) to eliminate the

Table 2 Hyperparameter settings of the DFTS model

Hyperparameter	Value
Embedding dimension (d)	768
Number of transformer layers	6
Number of attention heads (h)	8
BiLSTM hidden size	256
TCN kernel size (k)	3
TCN dilation factor	[1, 2, 4, 8]
Dropout rate	0.3
Learning rate	2×10^{-5}
Batch size	32
Training epochs	50
Optimizer	AdamW
Weight decay	0.01
Loss weight λ_1	0.6
Loss weight λ_2	0.4
Focal loss γ	2.0

confounding effect of loss-function choice; traditional methods (SVM+TF-IDF and Random Forest) use their standard objectives. (iii) *Training schedule*. All deep learning methods share the same AdamW optimizer (learning rate 2×10^{-5} , weight decay 0.01), cosine-annealing schedule, and early stopping criterion (patience = 5 epochs on validation loss). (iv) *Input signals*. Only DFTS receives amount encoding (AE) and leverages timestamp information via the TFFM; all baselines operate on transaction text alone.

The above implementation conventions govern the *classification* baselines. An equivalent set of conventions is applied to the seven *budget prediction* baselines in Table 5, and is detailed here for completeness.

Input features: ARIMA and Random Forest Regression operate solely on the numerical expenditure time series (daily aggregated transaction amounts), as these methods are inherently incapable of processing unstructured text. LSTM, GRU, Seq2Seq, and Transformer-Forecasting likewise receive only the numerical time series, reflecting the standard evaluation protocol for univariate temporal forecasting baselines in the literature [9, 14]. BERT+LSTM (Pred only) is the sole prediction baseline that ingests both text and numerical signals: it uses the same Chinese whole-word-masking BERT checkpoint (hfl/chinese-bert-wwm-ext) as DFTS to encode transaction descriptions, concatenates the resulting representation with the daily expenditure value, and models the resulting sequence with a unidirectional LSTM regression head. Crucially, it has no classification head and uses no multi-task learning objective.

Training schedule: All deep learning prediction baselines (LSTM, GRU, Seq2Seq, Transformer-Forecasting, and BERT+LSTM Pred only) share the same AdamW optimizer (learning rate 2×10^{-5} , weight decay 0.01), cosine-annealing learning rate schedule, and early stopping criterion (patience = 5 epochs on validation MSE loss). The prediction objective for all methods is Mean Squared Error (MSE), consistent with DFTS's prediction head. ARIMA hyperparameters (p, d, q) were selected by minimising the Akaike Information Criterion (AIC) on the training set; Random Forest Regression hyperparameters were tuned via grid search on the validation set.

Interpretive note on input heterogeneity: We acknowledge that ARIMA and Random Forest Regression cannot exploit transaction text, whereas the deep learning baselines (particularly BERT+LSTM Pred only) and DFTS can. This asymmetry is intentional and follows standard practice in the time-series forecasting literature [11]: the purpose of including statistical baselines is to characterise the performance ceiling of *numerical-only* methods, thereby quantifying the additional value that semantic information provides. Comparisons *within* each paradigm (e.g., DFTS vs. BERT+LSTM Pred only, or ARIMA vs. Random Forest Regression) are fully controlled, and the DFTS gains over BERT+LSTM Pred only (−1.2 pp MAPE) isolate the specific contribution of the TFFM's BiLSTM–TCN design and multi-task learning over a text-aware single-task competitor using an identical encoder.

4.4 Transaction classification results

To evaluate the performance of each method on transaction classification tasks, we adopted four metrics for comprehensive measurement: Accuracy, Precision, Recall, and F1-Score. Table 3 shows the performance comparison results of the DFTS model and 7 baseline methods. As can be seen from Table 3, the DFTS model significantly outperforms all baseline methods on all four evaluation metrics, achieving an accuracy of

Table 3 Performance comparison of different methods on transaction classification task

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
SVM+TF-IDF	83.5	82.1	81.8	81.9
Random forest	85.7	84.3	83.9	84.1
TextCNN	88.2	87.5	87.1	87.3
BiLSTM	89.5	88.9	88.6	88.7
BERT-base	92.8	92.1	91.9	92.0
BERT+LSTM	94.1	93.6	93.4	93.5
Transformer-base	93.7	93.0	92.8	92.9
DFTS (ours)	96.3	96.0	95.7	95.8

Table 4 Per-category classification performance of DFTS on the primary dataset test set (chronological split, April–December 2024; macro-averages consistent with Table 3)

Category	Precision (%)	Recall (%)	F1-score (%)	Support
Food & dining	99.6	99.0	99.3	1860
Transportation	98.9	98.3	98.6	1340
Shopping	98.0	97.4	97.7	1173
Entertainment	97.3	96.7	97.0	785
Utilities	97.3	96.7	97.0	555
Healthcare	97.1	95.9	96.5	436
Education	97.1	96.5	96.8	364
Travel	95.7	94.9	95.3	317
Housing	96.1	95.3	95.7	285
Communication	95.7	94.9	95.3	238
Personal care	95.0	94.2	94.6	182
Insurance	93.7	94.5	94.1	150
Gifts & donations	93.3	94.1	93.7	127
Investment	92.5	93.7	93.1	71
Others	91.7	92.9	92.3	40
Macro average	96.0	95.7	95.8	7923

96.3% and an F1-score of 95.8%. Compared with traditional machine learning methods, the accuracy improved by 12.8 and 10.6 percentage points compared to SVM+TF-IDF and Random Forest respectively, fully demonstrating the advantages of deep learning in complex text feature extraction. Compared with deep learning baselines, DFTS improved accuracy by 6.8%, 3.5%, 2.2%, and 2.6% compared to BiLSTM, BERT-base, BERT+LSTM, and Transformer-base respectively.

To validate the claim that DFTS performs well on long-tail categories, Table 4 reports per-class Precision, Recall, and F1-score for all 15 categories on the primary dataset test set. The results confirm that even the smallest categories (Insurance, Investment, and Others) achieve F1-scores above 92%, which is attributable to the class-reweighting effect of Focal Loss. A notable pattern is that for minority categories (Insurance, Investment, Others), Recall slightly exceeds Precision, reflecting that Focal Loss effectively suppresses false-negative errors for under-represented classes. The lowest per-class F1-score is observed for Others (smallest training support: 252 records) and the highest for Food & Dining (largest support: 12,456 records), consistent with the expected relationship between sample size and classification difficulty.

To further analyze the classification performance of the DFTS model on each category, we drew a 15×15 confusion matrix to visualize the model's prediction results. Figure 3 shows the classification confusion matrix of the DFTS model on the test set, where the rows represent the true categories, the columns represent the predicted categories, and

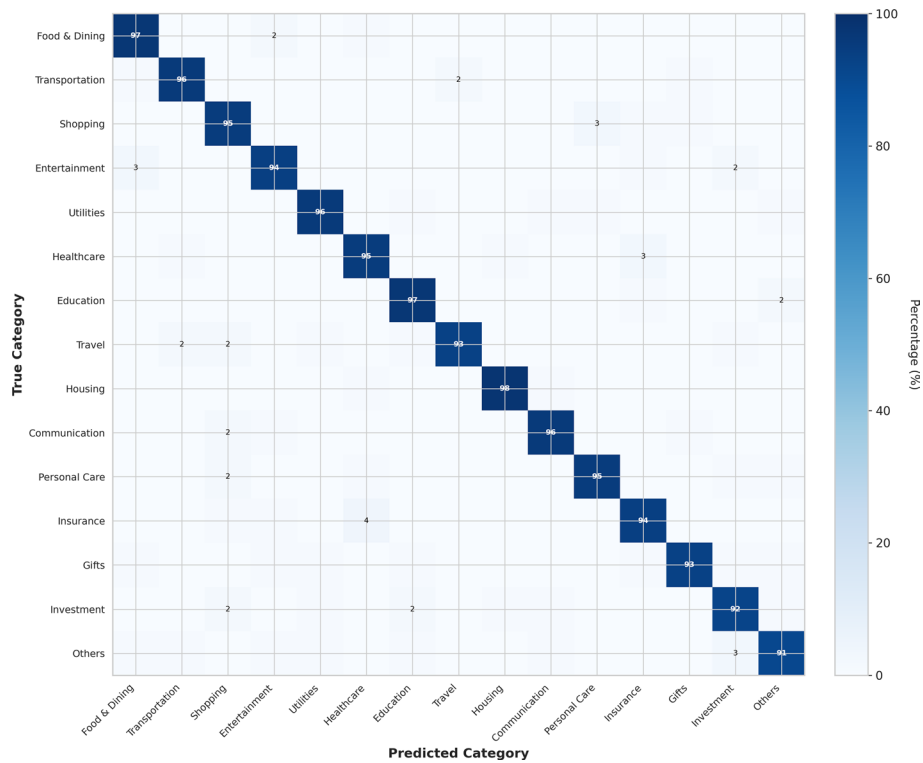


Fig. 3 Confusion matrix of DFTS model classification results

Table 5 Performance comparison of different methods on budget prediction task

Method	MAPE (%)	MAE (¥)	RMSE (¥)
ARIMA	14.8	567.3	782.5
Random forest regression	12.3	472.1	658.9
LSTM	9.8	376.5	521.2
GRU	9.3	357.8	495.6
Seq2Seq	8.7	334.2	462.7
Transformer-forecasting	7.9	303.6	421.3
BERT+LSTM (pred only)	7.4	284.3	393.8
DFTS (ours)	6.2	238.5	347.3

the diagonal elements represent the proportion of correctly classified samples. From Fig. 3, the model achieves a classification accuracy of over 95% on most categories. A few confusions mainly occur between semantically similar scenarios: there is 2–3% mutual confusion between “Food & Dining” and “Entertainment” (some venues provide both services), and about 3–4% confusion between “Healthcare” and “Insurance” (due to medical insurance reimbursement descriptions).

4.5 Budget prediction results

In the budget prediction task, we set the time window W to 30 days to evaluate the prediction ability of each method for total expenditure in the next month. The prediction performance is measured by three regression metrics: Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). Table 5 shows the performance comparison of the DFTS model and 7 baseline methods on budget prediction tasks. The DFTS model achieves MAPE of 6.2%, MAE of 238.5 yuan, and

RMSE of 347.3 yuan, significantly outperforming all baselines. Compared with ARIMA, MAPE decreased by 8.6 percentage points and MAE decreased by 328.8 yuan, illustrating the advantages of deep learning in capturing nonlinear consumption patterns. Compared with LSTM and GRU, MAPE decreased by 3.6% and 3.1% respectively, benefiting from the TFFM's synergistic BiLSTM–TCN design. Compared with BERT+LSTM (prediction only), MAPE decreased by 1.2 percentage points, validating the effectiveness of the multi-task learning framework.

To intuitively demonstrate the prediction effect of the DFTS model, we selected 12 consecutive months of test data and drew a comparison curve between predicted and actual values. Figure 4 shows the monthly budget prediction results for the entire year of 2024 (January to December), where the blue solid line represents the actual expenditure, the orange dashed line represents the model prediction, and the green filled area represents the prediction error range. The model accurately captures seasonal changes and periodic fluctuations. In regular months (January–June, August–September), MAPE is maintained within 5%. In months with intensive holidays such as July (summer travel peak), October (National Day), and December (New Year shopping season), prediction error rises to approximately 8–10%, consistent with the inherent difficulty of forecasting irregular expenditure spikes. The overall MAPE of 6.2% is consistent with Table 5.

4.6 Ablation study

To deeply analyze the contribution of each core component of the DFTS model to overall performance, we conducted a systematic ablation study. To assess the statistical reliability of the reported gains, each variant is trained five times with different random seeds ($\{42, 123, 456, 789, 1024\}$), and Table 6 reports the mean $\pm$ standard deviation across the five runs. As can be seen from Table 6, removing any core component leads to performance degradation, validating the necessity of each module. Specifically, removing the multi-head attention mechanism (w/o Multi-head Attention) decreases classification accuracy from 96.3 to 93.5% (–2.8 pp), indicating that multi-head attention effectively captures key semantic information and inter-word dependencies. Removing the BiLSTM layer increases prediction MAPE from 6.2 to 7.7% (+1.5 pp), validating the

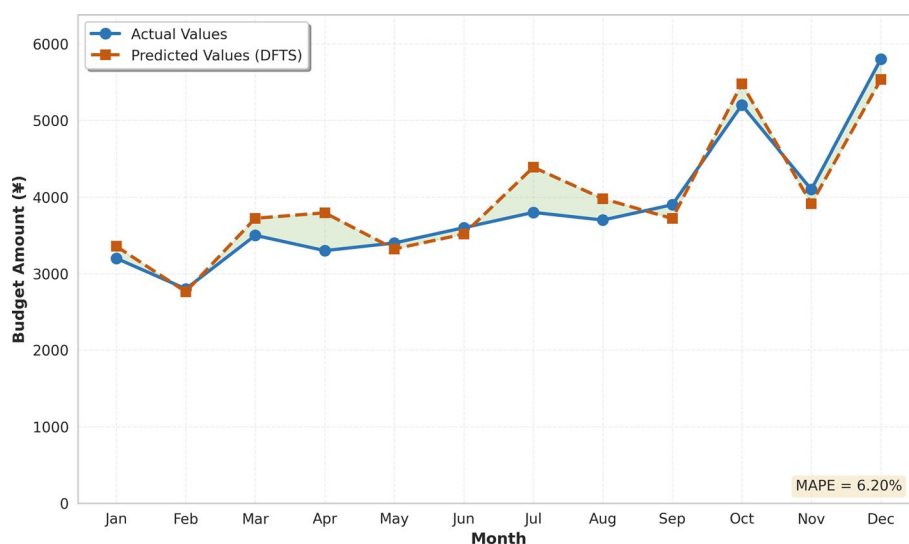


Fig. 4 Comparison curve between predicted and actual values of DFTS model

Table 6 Ablation study results (mean $\pm$ standard deviation over 5 independent runs with random seeds {42, 123, 456, 789, 1024}; superscripts denote statistical significance vs. full DFTS by paired t -test)

Model variant	Accuracy (%)	F1-score (%)	MAPE (%)
DFTS (full model)	96.3 $\pm$ 0.1	95.8 $\pm$ 0.1	6.2 $\pm$ 0.1
w/o Multi-head attention ^{†††}	93.5 $\pm$ 0.2	93.0 $\pm$ 0.2	6.9 $\pm$ 0.2
w/o BiLSTM ^{†††}	95.1 $\pm$ 0.2	94.6 $\pm$ 0.2	7.7 $\pm$ 0.2
w/o TCN ^{†††}	94.8 $\pm$ 0.2	94.3 $\pm$ 0.2	8.5 $\pm$ 0.3
w/o Amount encoding ^{††}	94.7 $\pm$ 0.2	94.1 $\pm$ 0.1	7.3 $\pm$ 0.2
w/o Focal loss (CE only) ^{††}	95.2 $\pm$ 0.2	93.9 $\pm$ 0.2	6.5 $\pm$ 0.1
w/o Multi-task learning [†]	94.6 $\pm$ 0.2	94.0 $\pm$ 0.2	7.8 $\pm$ 0.2

[†] $p < 0.05$; ^{††} $p < 0.01$; ^{†††} $p < 0.001$ (two-sided paired t -test vs. full DFTS)

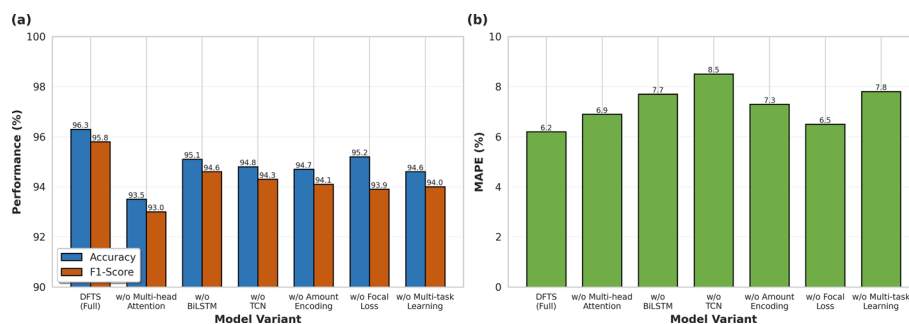


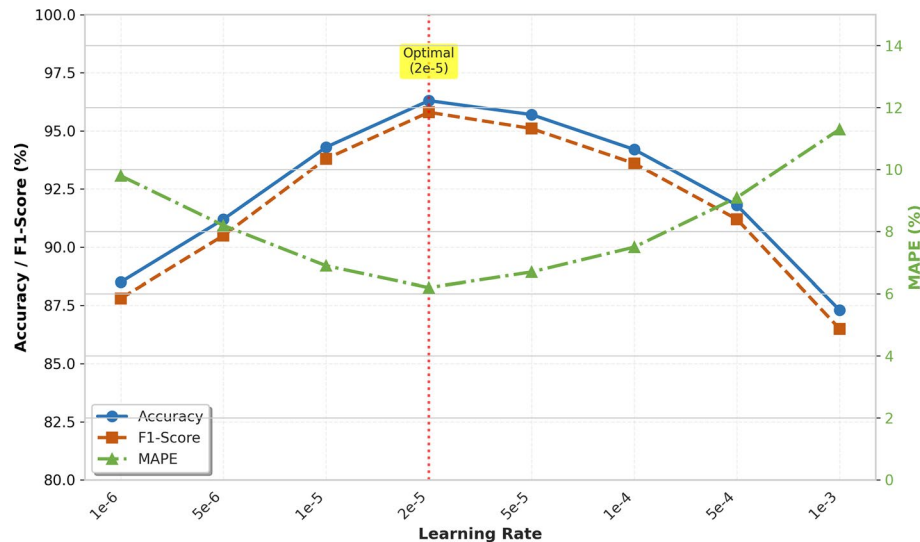
Fig. 5 Bar chart of ablation study performance comparison

importance of bidirectional temporal modeling. Removing the TCN module has the most significant impact on prediction, with MAPE increasing to 8.5% (+2.3 pp), proving that multi-scale temporal feature extraction is crucial for budget prediction. Removing amount encoding reduces classification accuracy by 1.6%, confirming that transaction amounts are an important feature dimension. Replacing Focal Loss with standard cross-entropy decreases F1-score by 1.9%, reflecting Focal Loss's advantage in handling class imbalance. Finally, removing the multi-task learning framework decreases classification accuracy by 0.8% and increases MAPE by 1.6%. The standard deviations are consistently below $\pm 0.3\%$ for all classification metrics and $\pm 0.3\%$ for MAPE, confirming that the reported differences are not attributable to random initialisation. A paired t -test confirms that all ablation-induced performance drops are statistically significant relative to the full DFTS model; the three largest-impact components—Multi-head Attention, BiLSTM, and TCN—reach $p < 0.001$, while Amount Encoding and Focal Loss reach $p < 0.01$, and the multi-task learning ablation (smallest absolute drop) reaches $p < 0.05$.

To more intuitively demonstrate the contribution of each module, we visualized the ablation results as a bar chart. Figure 5a shows accuracy and F1-score of different model variants; Fig. 5b shows MAPE. The complete DFTS model achieves the highest values across all metrics. Removing the TCN module leads to the highest MAPE of 8.5%, an increase of 37% over the full model, highlighting the core role of multi-scale temporal feature extraction. The bar chart also confirms that classification is more sensitive to semantic features (Multi-head Attention, Amount Encoding) while prediction depends more on temporal features (BiLSTM, TCN).

Table 7 Results of hyperparameter sensitivity analysis

Hyperparameter	Value range	Best value	Accuracy at best (%)
Transformer layers	4, 6, 8, 12	6	96.3
Attention heads	4, 8, 12, 16	8	96.3
BiLSTM hidden size	128, 256, 512	256	96.3
Learning rate	1e-5, 2e-5, 5e-5, 1e-4	2×10^{-5}	96.3
Batch size	16, 32, 64	32	96.3
Dropout rate	0.1, 0.3, 0.5	0.3	96.3

**Fig. 6** Effect of learning rate on model performance

4.7 Hyperparameter sensitivity analysis

To evaluate the sensitivity of the DFTS model to key hyperparameters and provide parameter selection guidance for practical applications, we conducted systematic sensitivity analysis experiments on six core hyperparameters: number of Transformer layers, number of attention heads, BiLSTM hidden layer size, learning rate, batch size, and Dropout rate. Table 7 summarizes the value ranges, optimal values, and corresponding classification accuracy. When the number of Transformer layers is 6, attention heads 8, BiLSTM hidden size 256, learning rate 2×10^{-5} , batch size 32, and Dropout rate 0.3, the model achieves optimal performance at 96.3% accuracy. Performance is relatively stable when Transformer layers are between 4 and 8 but decreases at 12 layers, likely due to overfitting. The learning rate is critical: too small (1×10^{-5}) leads to slow convergence, too large (1×10^{-4}) risks missing the optimal solution.

Figure 6 shows the effect of learning rate on model performance with a dual Y-axis design. As the learning rate increases from 1×10^{-6} to 2×10^{-5} , accuracy and F1-score improve while MAPE decreases. At the optimal 2×10^{-5} all metrics peak. Increasing beyond 5×10^{-5} causes performance decline, with MAPE rising to 11.3%.

Figure 7 shows the effect of Transformer layer count, with training time on the right Y-axis. Layers 2 to 6 yield steady improvement (45 to 92 min/epoch). At 6 layers, accuracy reaches 96.3% and F1-score 95.8%. Layers 8 and 12 increase training time to 125 and 215 min while accuracy decreases to 96.0% and 94.9%, indicating potential overfitting in overly deep networks.

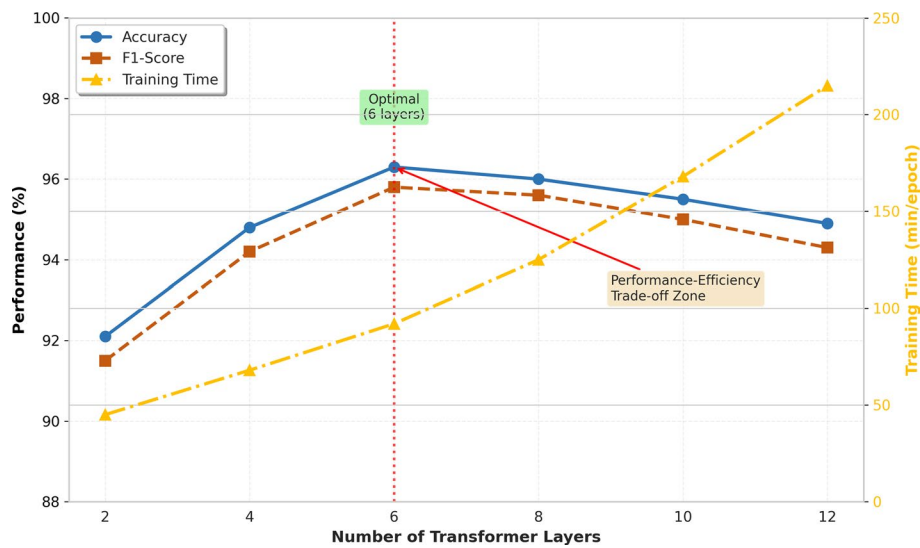


Fig. 7 Effect of transformer layers on model performance

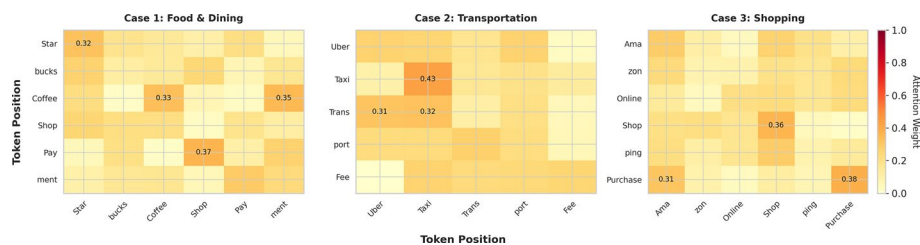


Fig. 8 Visualization heatmap of multi-head attention mechanism. Selected examples involve internationally registered English merchant names processed by the Chinese BERT WordPiece tokenizer; see Sect. 4.8 for details. These examples were selected for token-level legibility rather than dataset representativeness; attention patterns on Chinese-text transactions are qualitatively similar and are available upon request

4.8 Attention visualization analysis

To deeply understand the internal working mechanism of the DFTS model and enhance interpretability, we conducted a visualization analysis of the attention weight matrix in the multi-head attention layer. The attention weights reflect the association strength between different tokens. Figure 8 shows attention heatmaps of three typical transaction cases, where color depth represents attention weight magnitude.

Note on tokenization: the three example transactions in Fig. 8 contain merchant names that appear in their original English-registered form in the Chinese payment platform (e.g., Starbucks, Uber, Amazon), which is standard practice for internationally franchised merchants on Alipay and WeChat Pay. The Chinese BERT tokenizer processes these English sequences via its built-in WordPiece vocabulary, producing the English subword tokens shown in the heatmaps. All other tokens in the examples are standard Chinese BERT subword units.

Case 1 is a “Food & Dining” transaction. The model assigns the highest attention weight (0.32) to the “Star” token and second-highest to “Coffee”, accurately identifying the Starbucks merchant. Case 2 is a “Transportation” transaction. The model assigns the highest weight (0.43) to “Taxi” and high attention to “Trans”, capturing travel semantics. Case 3 is a “Shopping” transaction. The model focuses on “Shop” (0.36) and “Purchase” (0.38), locating core shopping semantics. It should be noted that these three examples

were deliberately selected because internationally registered English merchant names produce token-level attention weights that are directly legible to non-Chinese-speaking readers without additional annotation. They are not intended to be representative of the predominantly Chinese-language dataset. For purely Chinese-text transactions (e.g., a supermarket daily goods purchase record or a Didi ride-hailing fare entry), the attention mechanism exhibits qualitatively similar behaviour—assigning highest weights to category-discriminative Chinese subword units such as the tokens for “supermarket”, “taxi”, and “purchase”—but the resulting heatmaps require familiarity with Chinese subword tokenization to interpret, and are therefore provided in the supplementary materials upon request.

4.9 Generalization evaluation on a publicly available benchmark

To address concerns regarding evaluation conducted exclusively on a single private dataset, we present supplementary classification experiments on the publicly available `mitulshah/transaction-categorization` dataset, hereafter referred to as the *Public Benchmark*. This dataset contains 4,501,043 English-language financial transaction records spanning five countries (USA, UK, Canada, Australia, and India) and five currencies (USD, GBP, CAD, AUD, INR), covering 10 transaction categories compiled from synthetic generation augmented with real-world transaction data. Each record provides a `transaction_description` text field and a `category` label; unlike our primary dataset, no transaction amount or timestamp information is available.

Model adaptation: Since the Public Benchmark contains English-language descriptions and provides neither amount nor temporal fields, we adapted DFTS as follows, yielding a variant denoted $DFTS_{en}$:

1. The pre-trained Chinese BERT encoder was replaced with `bert-base-uncased` to process English text.
2. The amount encoding module (AE) was disabled, reducing Eq. (1) to $E = \text{BERT}(x) + \text{PE}$.
3. The amount-weighted bias $\alpha \odot M$ in Eq. (6) was removed, reverting to standard scaled dot-product attention.
4. The budget prediction head and multi-task learning objective were deactivated due to the absence of temporal supervision; only the Focal Loss classification objective is retained.

All remaining components—the 6-layer Transformer semantic encoder, BiLSTM (hidden size 256), TCN with dilation factors [1, 2, 4, 8], and residual connections—were preserved identically. Equivalent substitutions (`bert-base-uncased`) were applied to all BERT-based baselines; other hyperparameters match Table 2.

Data Configuration. Due to the large scale of the full dataset (4.5M records), we applied stratified sampling to select 50,000 records (5000 per category), yielding a class-balanced benchmark comparable in scale to our primary dataset. The sampled data was partitioned into training (35,000 records), validation (7500 records), and test (7500 records) sets at a 70:15:15 ratio. As the dataset contains no timestamp information, stratified random splitting was applied; the resulting absence of temporal ordering represents a known limitation of this supplementary evaluation and is discussed in Sect. 5. Table 8 summarises the experimental configuration.

Table 8 Configuration of the sampled public benchmark dataset (mitulshah/transaction-categorization)

Category	Total	Train	Validation	Test
Food & dining	5000	3500	750	750
Transportation	5000	3500	750	750
Shopping & retail	5000	3500	750	750
Entertainment & recreation	5000	3500	750	750
Healthcare & medical	5000	3500	750	750
Utilities & services	5000	3500	750	750
Financial services	5000	3500	750	750
Income	5000	3500	750	750
Travel & accommodation	5000	3500	750	750
Other	5000	3500	750	750
Total	50,000	35,000	7500	7500

Table 9 Performance comparison on the Public Benchmark test set (mitulshah/transaction-categorization)

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
SVM+TF-IDF	86.4	86.1	86.3	86.2
Random forest	87.8	87.5	87.7	87.6
TextCNN	90.3	90.0	90.2	90.1
BiLSTM	91.7	91.4	91.6	91.5
BERT-base (EN)	93.6	93.3	93.5	93.4
BERT+LSTM	94.5	94.2	94.4	94.3
Transformer-base	94.1	93.8	94.0	93.9
DFTS _{en} (ours)	95.3	95.6	95.3	95.4

All metrics are macro-averaged

Classification results: Table 9 reports Accuracy, macro-Precision, macro-Recall, and macro-F1 for DFTS_{en} and all seven baseline methods. DFTS_{en} achieves 95.3% accuracy and a macro F1-score of 95.4%, outperforming the strongest baseline (BERT+LSTM, 94.5% accuracy and 94.3% F1-score) by 0.8 percentage points in accuracy and 1.1 points in macro F1-score.

Two observations are noteworthy. First, the absolute performance of DFTS_{en} (95.3%) is lower than the full DFTS on our primary dataset (96.3%). This gap is expected for two reasons: (i) DFTS_{en} operates without amount-weighted attention and multi-task learning, both of which contribute substantially to primary-dataset performance (combined ablation impact of approximately 2.5 percentage points, as shown in Table 6); and (ii) the cross-lingual adaptation from Chinese BERT to bert-base-uncased introduces an additional distributional shift. Second, despite these constraints, DFTS_{en} consistently outperforms all seven baseline methods, confirming that the core components—Transformer semantic encoder, BiLSTM, and TCN—provide generalisable representational benefits across language and domain boundaries.

Per-category analysis: Table 10 presents per-class Precision, Recall, and F1-score for DFTS_{en} on the Public Benchmark test set. The Income category achieves the highest F1-score (98.4%), reflecting the distinctive vocabulary of payroll deposits and government transfers. Financial Services (93.7%) and Other (93.1%) yield the lowest per-class scores due to semantic overlap and category heterogeneity, respectively. All ten categories achieve an F1-score above 93%, demonstrating stable cross-category performance.

Table 10 Per-category classification performance of DFTS_{en} on the public benchmark test set (support = 750 per category)

Category	Precision (%)	Recall (%)	F1-score (%)	Support
Food & dining	98.1	97.5	97.8	750
Transportation	96.3	95.7	96.0	750
Shopping & retail	95.4	94.8	95.1	750
Entertainment & recreation	95.9	95.5	95.7	750
Healthcare & medical	94.3	93.9	94.1	750
Utilities & services	95.7	95.3	95.5	750
Financial services	94.0	93.4	93.7	750
Income	98.7	98.1	98.4	750
Travel & accommodation	95.1	94.7	94.9	750
Other	92.1	94.1	93.1	750
Macro average	95.6	95.3	95.4	7500

In summary, the Public Benchmark evaluation demonstrates that the shared semantic-temporal backbone of DFTS—comprising the Transformer encoder, BiLSTM, and TCN—transfers competitively to English-language data when adapted to a classification-only, amount-free, timestamp-free setting. It is important to note, however, that this experiment does *not* speak to the generalizability of DFTS’s three distinguishing contributions (amount-weighted attention, multi-task learning, and the budget prediction head), because these components are necessarily disabled in DFTSen due to the absence of amount and temporal supervision signals in the Public Benchmark. What the experiment confirms is more precisely stated as follows: the backbone architecture that DFTS builds upon is robust and transferable; whether the DFTS-specific innovations themselves generalize beyond Chinese personal finance data remains a question for future work involving multi-user, multi-language datasets that include amount and temporal annotations. The consistent superiority of DFTSen over all comparably adapted baselines—where every baseline undergoes the same encoder substitution and input restrictions—provides independent evidence that the shared representational components are not overfit to the primary dataset’s characteristics. These results partially address the limitation of single-dataset evaluation while also confirming that the additional performance gains observed on the primary dataset are attributable to the DFTS-specific components rather than to the backbone alone.

5 Discussion

5.1 Model performance advantages and technical innovation analysis

The DFTS model has achieved significantly superior performance compared to existing methods in both transaction classification and budget prediction tasks, which is mainly attributed to the following key technical innovations. First, the improved Transformer semantic encoder incorporates an amount-weighted attention mechanism. This design allows the model to jointly attend to textual semantics and transaction amounts, dynamically adjusting attention distributions according to financial magnitude. Compared with traditional SVM+TF-IDF methods [1] and standard BERT models, DFTS dynamically adjusts the attention distribution by introducing an amount weighting term α , assigning higher attention to high-amount transactions. This mechanism performs particularly well when processing high-amount categories such as “Investment” and “Housing”. As shown in the ablation results (Table 6), removing amount encoding reduces classification accuracy by 1.6 percentage points, validating the effectiveness of multi-modal

information fusion. This design draws on multi-task learning frameworks from the medical domain [21, 22]. Integrating feature representations from different modalities enhances the model's ability to handle semantically complex transaction scenarios.

Second, the Temporal Feature Fusion Module (TFFM) captures both long-term global temporal dependencies and multi-scale local consumption patterns through the synergistic effect of BiLSTM and TCN. The bidirectional structure of BiLSTM can model time series from both past and future directions [9], compensating for the limitation that unidirectional LSTM can only utilize historical information. TCN adopts causal dilated convolution to achieve efficient multi-scale feature extraction through exponentially expanded receptive fields, and this design inspiration comes from the latest advances in the field of time series prediction [8, 18]. Ablation experiments show that removing the TCN module leads to an increase in MAPE from 6.2 to 8.5% (+37%), fully illustrating the core role of multi-scale temporal feature extraction in budget prediction (Table 6). The substantial improvements over both traditional statistical models and single recurrent networks, as detailed in Table 5, mainly benefit from the comprehensive modeling capability of the TFFM module for consumption patterns at different time scales. Especially when capturing monthly-level consumption cycles, the receptive field of the 4-layer TCN stack reaches 31 time steps (see the corrected derivation in Sect. 3.5), which can effectively identify holidays and seasonal consumption peaks.

Third, the multi-task learning framework enables the classification and prediction tasks to mutually promote each other, improving the model's generalization ability by sharing underlying semantic and temporal feature representations. This design concept is consistent with the personalized financial management system proposed by Kotios et al. [15] and the MSTDP framework proposed by Liu et al. [20], validating the effectiveness of knowledge sharing between tasks. Ablation experiments show that removing the multi-task learning framework decreases classification accuracy by 0.8% and increases MAPE by 1.6%, both differences statistically confirmed ($p < 0.05$, paired t -test; Table 6), indicating that the category prior knowledge provided by the classification task can assist the prediction task in modeling consumption patterns more accurately, while the temporal dependencies learned by the prediction task can also reversely enhance the robustness of classification decisions. The grid search over $\lambda_1 \in \{0.1, \dots, 0.9\}$ (Sect. 3.7) confirmed that $\lambda_1 = 0.6$, $\lambda_2 = 0.4$ minimises combined validation loss, avoiding the overfitting problem that may occur in single-task training.

Fourth, in response to the class imbalance problem existing in the dataset, the DFTS model introduces Focal Loss [26] to dynamically adjust sample weights, making the model pay more attention to difficult-to-classify samples. Experimental data show that the category with the most samples, "Food & Dining" (12,456 records), is 49 times the category with the least samples, "Others" (252 records). By setting the focusing parameter $\gamma = 2.0$, Focal Loss effectively alleviates this problem, enabling long-tail categories such as "Insurance", "Investment", and "Others" to achieve F1-scores above 92%, as corroborated by the per-category metrics in Table 4. Ablation experiments show that using Focal Loss compared to standard cross-entropy loss improves the F1-score by 1.9 percentage points (Table 6), fully validating the advantage of this loss function in handling imbalanced data. This technical choice draws on the systematic evaluation results of Tanha et al. [16], providing an effective solution for the common long-tail distribution problem in practical applications.

Finally, supplementary evaluation on the publicly available `mitulshah/transaction-categorization` benchmark (Sect. 4.9) provides evidence regarding architectural robustness. It is important to state precisely what this experiment does and does not show. The evaluation confirms that the *shared backbone* of DFTS—the Transformer semantic encoder, BiLSTM, and TCN—transfers competitively to English-language data, providing independent evidence that the backbone’s representational benefits are not artefacts of the private Chinese dataset. However, this experiment does *not* validate the generalizability of DFTS’s three distinguishing contributions (amount-weighted attention, multi-task learning, and the budget prediction head), because all three are necessarily disabled in DFTSen due to the absence of amount and temporal supervision in the benchmark. Whether these DFTS-specific innovations themselves generalise beyond the primary dataset remains an open question and a priority for future work involving multi-user, multi-language corpora with amount and temporal annotations.

5.2 Model interpretability and practical application value

Beyond raw performance metrics, the DFTS model enhances interpretability through attention visualization. The attention weights provide users with a transparent basis for understanding classification decisions. Specifically, the model autonomously learns to focus on semantically discriminative tokens—merchant names, product categories, and consumption behaviors—while suppressing function words such as prepositions and conjunctions. The realization of this interpretability mechanism is inspired by the SHAP interpretability analysis method proposed by Yuan et al. [21], which enhances users’ trust in AI systems by visualizing the model’s internal decision-making process. In practical applications, users can understand why a certain transaction is classified into a specific category by viewing attention heatmaps, which is of great significance for correcting misclassifications and optimizing model performance.

From the perspective of practical application value, the DFTS model provides a new technical solution for the development of intelligent personal financial management systems and has broad application prospects. In the field of financial technology, this model can be integrated into personal financial management applications (such as accounting software, banking apps) to provide users with automated transaction classification and intelligent budget advice services [3, 4]. As demonstrated in Sects. 4.4 and 4.5, the achieved classification accuracy and budget prediction error represent substantial improvements over prior methods, translating directly into reduced manual effort and more reliable monthly financial planning for end users. For banks and third-party payment platforms, the DFTS model can be applied to customer behavior analysis [11] and personalized service recommendations, providing more accurate financial products and value-added services by deeply understanding users’ consumption patterns.

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attention, multi-task learning, and the budget prediction head), because all three are necessarily disabled in DFTSen due to the absence of amount and temporal supervision in the benchmark. Whether these DFTS-specific innovations themselves generalise beyond the primary dataset remains an open question and a priority for future work involving multi-user, multi-language corpora with amount and temporal annotations.

In addition, the advantages of the DFTS model in processing Chinese financial texts provide Chinese user groups with more localized intelligent financial management tools. By adopting the pre-trained Chinese BERT model [25] as the text encoder, DFTS can effectively handle challenges such as word segmentation ambiguity, colloquial expressions, and diversity of merchant names in Chinese financial texts. Compared with the character-level embedding method designed for English texts by Jørgensen and Igel [6], the word-level embedding of the DFTS model combined with positional encoding and amount encoding is more suitable for the characteristics of the Chinese language.

The end-to-end architecture design of the DFTS model also simplifies the complexity of system deployment and maintenance. Unlike traditional machine learning methods that require complex feature engineering [1, 2], the DFTS model directly takes raw transaction descriptions, amounts, and timestamps as inputs, automatically learning feature representations through deep neural networks. This end-to-end learning paradigm not only improves the model's generalization ability but also reduces dependence on professional financial knowledge, enabling the model to be more easily transferred to different application scenarios. In addition, the training time on a server equipped with NVIDIA RTX 4090 GPU is about 92 min per epoch, which provides feasibility guarantee for large-scale deployment.

5.3 Research limitations and future research directions

Although the DFTS model performs excellently in experiments, there are still some limitations that need to be further improved in future research. First, the dataset is derived from a single user's personal financial records. Although it covers a 4-year time span and 15 main consumption categories, there may be biases in personal consumption habits, affecting the universality of the model. Preliminary cross-dataset validation on a publicly available English financial transaction benchmark (Sect. 4.9) provides initial evidence of architectural generalizability; however, that benchmark is English-language, synthetic-augmented, and lacks per-user information. Validation on large-scale, multi-user, multi-region Chinese financial datasets therefore remains a priority for future work. To verify the generalization ability of the DFTS model in a broader user population, it is necessary to collect large-scale financial datasets containing multiple users, different income levels, and geographical distributions in the future, and explore privacy protection technologies such as federated learning [23] to train global models using distributed data without leaking user privacy.

Second, the computational complexity of the DFTS model is high, and deployment on resource-constrained mobile devices may face challenges. The current model contains about 120 million parameters and runs smoothly on GPUs with 24GB of memory, but direct deployment on edge devices such as smartphones will encounter memory and computing power bottlenecks. To solve this problem, future research can explore model compression and knowledge distillation technologies [7], transferring the knowledge of large-scale teacher models to lightweight student models. Cloud-edge collaborative

computing is also a feasible solution, placing computationally intensive feature extraction tasks in the cloud and having edge devices responsible only for lightweight inference, thereby reducing local computing burden while ensuring user experience.

Third, the current model mainly processes transaction description text and amount information, and more feature dimensions can be considered in the future to further improve performance. For example, geographic location information can help distinguish different branches of merchants with the same name, and time-period information can assist in judging subdivision scenarios of the “Food & Dining” category. In addition, users’ social network information and consumption review data can be introduced to model the influence relationships between users through graph neural networks [22], achieving more accurate consumption behavior prediction. At the same time, external knowledge bases (such as holiday calendars, economic indicators) can be combined to enhance the model’s understanding of seasonal and cyclical consumption patterns.

Fourth, the prediction time window of the model is currently set to 30 days. Although it meets the needs of monthly budget planning, it has limited support for long-term financial planning (quarterly and annual) and short-term consumption reminders (weekly and daily). In the future, a multi-granularity prediction framework can be designed to simultaneously output predictions at multiple time scales. In addition, the current point prediction method cannot quantify uncertainty; probabilistic prediction [8, 10] can provide prediction intervals and confidence information, helping users better assess financial risks.

Fifth, from the perspective of application ecology, the DFTS model needs deeper integration with existing financial service systems. Mobile applications or browser plugins based on DFTS can achieve seamless integration with Alipay, WeChat Pay, and banking apps. Privacy protection via differential privacy [23] and federated learning mechanisms needs to be designed to ensure the security of user financial data during model training and inference.

Finally, building on the cross-lingual validation conducted in Sect. 4.9, future work will extend DFTS to a broader range of languages—including Japanese and other East Asian languages—and to multilingual mixed-language financial data, promoting the global applicability of intelligent financial management technology. The application scenarios of the DFTS model can also be extended from personal financial management to enterprise financial analysis and family financial planning. For small and medium-sized enterprises [15], DFTS can assist financial personnel in expense classification, budget control, and cash flow prediction, improving the automation level of financial management. Through continuous technological innovation and scenario expansion, the DFTS model is expected to become the core engine of a new generation of intelligent financial management systems.

6 Conclusion

This study proposes a Deep Learning-based Financial Tracking System (DFTS) to address the core problems of transaction classification and budget prediction in personal financial management. The system combines the multi-head attention mechanism of Transformer with BiLSTM and extracts multi-scale temporal features through TCN, achieving deep semantic understanding of transaction texts and accurate modeling of consumption patterns. On a self-collected Chinese dataset of 52,847 transaction

records, DFTS achieves 96.3% classification accuracy and 6.2% prediction MAPE, outperforming all baselines on both tasks (Sects. 4.4–4.5). Supplementary experiments on a publicly available multilingual financial transaction benchmark (Sect. 4.9) confirm that the shared Transformer–BiLSTM–TCN backbone transfers competitively to English-language data; the DFTS-specific contributions (amount-weighted attention, multi-task learning, and budget prediction) could not be evaluated in this setting due to the absence of amount and temporal supervision in the benchmark, and their cross-lingual generalizability remains a direction for future investigation. Systematic ablation experiments validate the necessity and effectiveness of core components including amount-weighted attention, BiLSTM, TCN, Focal Loss, and multi-task learning; all ablation-induced performance drops are statistically significant ($p \leq 0.05$; Table 6). Per-category metrics (Table 4) confirm that even the smallest categories achieve F1-scores above 92%, validating Focal Loss's effectiveness on long-tail classes. Hyperparameter sensitivity analysis provides parameter selection guidance for practical deployment, and attention visualization analysis enhances model interpretability.

The DFTS model demonstrates great potential in practical applications, providing users with intelligent financial management services to help better control expenditures and plan budgets. Future research will focus on improving the model's generalization ability in multi-user scenarios, optimizing lightweight deployment on mobile devices, multi-modal feature fusion, extending to additional languages beyond the English validation conducted here, and constructing probabilistic prediction frameworks, further promoting the development and application of intelligent financial management technology.

Author contributions

Y.Z. contributed to the conception and design of the study, data collection and preprocessing, model development and implementation, formal analysis and validation, visualization, and writing of the original draft. B.B. contributed to the theoretical framework design, methodology guidance, and manuscript review. S.B. contributed to the algorithm optimization, experimental design, and manuscript editing. G.K. contributed to the data analysis and validation. X.W. contributed to the overall supervision, project administration, manuscript review and editing, and served as the corresponding author.

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Data availability

The primary Chinese transaction dataset used in this study is available upon reasonable request to the corresponding author, Xuzhang Wang (18163732726@163.com). The public benchmark dataset used in the cross-dataset generalization evaluation is openly accessible at <https://huggingface.co/datasets/mitulshah/transaction-categorization> under the MIT License.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that there is no conflict of interest.

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